

Muyi Bao

Phone: +1 412-512-8085

Email: muyib@andrew.cmu.edu

Website | [Google Scholar](#)

Education

Carnegie Mellon University (CMU)

M.S. in Electrical and Computer Engineering (Current)

- Cumulative GPA: 4.0/4.0

Pittsburgh, Pennsylvania, United States

Sep. 2025–June 2027 (Expected)

Xi'an Jiaotong-Liverpool University (XJTLU)

Bachelor of Engineering in Computer Science and Technology

- Cumulative GPA: 3.92/4.0 (First Class Honors)
- Honors: Academic Excellent Awards (Top 5%, 2023–2025 academic years)

Suzhou, Jiangsu, China

Sep. 2021–June 2025

Research Interests

My research focuses on building vision-language agents that can understand human instructions, interpret visual observations, and produce semantically aligned and physically executable behaviors in embodied environments. In particular, I am interested in developing model architectures and learning algorithms that improve the performance, efficiency, reliability, and safety of embodied agents as tasks, environments, sensory inputs, and robot embodiments become increasingly complex.

Research Experience

Graduate Research

SemPathBench: Semantic-Constrained Route Planning for Embodied Agents

Advisor: Prof. Ji Zhang, Carnegie Mellon University

Jul. 2026–Present

- Developing SemPathBench, a benchmark for language-guided route planning over known occupancy, object, and room maps, emphasizing complex grounding of relational, comparative, and multi-stage instructions together with soft constraints that require paths to optimize semantic spatial preferences, such as remaining closer to or farther from specified objects.
- Designed evaluation metrics that measure correct grounding and logical execution of hard constraints in multi-stage instructions, alignment with soft semantic preferences, path efficiency and overall success score.
- Implemented seven baseline planning methods for systematic comparison; their Overall Success Scores ranged from 0.01 to 0.18, highlighting the challenges of accurately grounding complex instructions and generating paths that align with soft constraints.

Goal2Pixel: Grounding Goals to Pixels for Vision-Language Navigation.

Advisors: Prof. Ji Zhang and Prof. Wenshan Wang, Carnegie Mellon University

Dec. 2025–Jun. 2026

- Proposed Goal2Pixel, a pure pixel-based paradigm that reformulates VLN-CE from low-level action prediction into navigable pixel grounding, achieving 54.1% SR on R2R-CE with approximately $6\times$ fewer VLM calls than direct action prediction.
- Developed Visibility-Aware Keyframe Memory (ViKeyMem), which selects compact and informative historical observations based on visibility changes and trajectory overlays, outperforming conventional history representations by 9% SR on average while reducing training and inference time by nearly 50%.
- Introduced semantic directive embeddings and coordinate-aware auxiliary losses to adapt pretrained vision-language models to image-space navigation, improving their understanding of action regions and sensitivity to pixel coordinates.

IntentNav: Learning Spatial-Visual Object Navigation from Human Demonstrations

Advisors: Prof. Ji Zhang and Prof. Wenshan Wang, Carnegie Mellon University

Dec. 2025–Jun. 2026

- Proposed IntentNav, a BEV-grounded spatial-visual imitation framework for learning human-like ObjectNav policies. Trained only on MP3D demonstrations, IntentNav achieved state-of-the-art performance among learning-based methods, reaching 53.8%, 70.5%, and 82.2% SR on MP3D, HM3D-v1, and HM3D-v2, respectively.
- Introduced Frontier-based Human-Intent Labeling to recover high-level search intent from low-level human actions, converting 23,767 demonstrations into 2.36 million candidate-level waypoint supervision samples and improving MP3D SR from 42.0% to 53.8% over trajectory-matching supervision.
- Designed a candidate-level VLM policy that integrates persistent BEV memory, candidate-aligned visual observations, agent-centric geometry, and an Intent-Aligned Objective to unify frontier exploration and target commitment. The same 2B VLM transferred zero-shot to wheeled, quadruped, and humanoid robots without additional fine-tuning.

Undergraduate Research

Vision Mamba in Remote Sensing: A Comprehensive Survey of Techniques, Applications and Outlook

Advisor: Prof. Guangliang Cheng, University of Liverpool

Feb. 2025–May 2025

- Conducted a systematic review of 120+ Mamba-based remote-sensing studies and established a taxonomy covering micro-level mechanisms, macro-level architectures, and downstream applications.
- Micro-architecture: Surveyed SSM formulations, scan strategies and multimodal/bi-temporal interaction mechanisms.
- Macro-architecture: Surveyed hybrid CNN/Transformer/Mamba designs, framework substitutions and frequency operations.
- Synthesized benchmark results across six downstream tasks, including semantic segmentation, object detection, and change detection, and identified seven future research directions, such as causality mitigation and computational efficiency.

FTCFormer: Fuzzy Token Clustering Transformer for Image Classification

Advisor: Prof. Wei Wang, XJTU

Jan. 2025–May 2025

- Proposed FTCFormer, a Transformer architecture with fuzzy token-clustering downsampling layers, and evaluated it on 32 datasets. Compared with the baseline, FTCFormer improved average accuracy by 1.43% on 5 fine-grained datasets, 1.09% on 6 natural datasets, 0.97% on 3 medical datasets, and 0.55% on 4 remote-sensing datasets.
- Designed Fuzzy Token Clustering and Merging (FTCM) using DPC-FKNN for clustering-center selection. DPC-FKNN combines density-peak clustering with fuzzy K-nearest-neighbor weighting to select more representative clustering centers.
- Introduced Spatial Connectivity Score for spatially aware token assignment and Channel Merging for channel-level token aggregation, preserving fine-grained semantic information beyond conventional token-level merging.

ASP-VMUNet: Atrous Shifted Parallel Vision Mamba U-Net for Skin Lesion Segmentation

Advisor: Prof. Guangliang Cheng, University of Liverpool

June 2024–Oct. 2024

- Proposed a CNN–Mamba hybrid U-Net architecture for skin-lesion segmentation, achieving state-of-the-art performance on ISIC16/17/18 and PH2.
- Designed Atrous Scan strategy to reduce background interference and expand Mamba’s receptive field, improving mIoU by 0.60%, 0.51%, 0.66%, and 0.49% on PH2 and ISIC16/17/18, respectively.
- Introduced Shift Round for parameter-free feature interaction and an SK-Net branch for dynamic local-global feature fusion. The SK-Net branch outperformed direct addition by 1.01%, 0.65%, 1.40%, and 0.39% on PH2 and ISIC16/17/18, respectively.

Publications ^{*}Equal contribution.

-
- [1] **Muyi Bao**^{*}, Yuxin Cai^{*}, Hang Xu, Zongtai Li, Jinxi He, Jingfan Tang, Chen Lv, Ji Zhang, Yaqi Xie, and Wenshan Wang. “Goal2Pixel: Grounding Goals to Pixels for Vision-Language Navigation.” *arXiv preprint*, 2026. Submitted to CoRL 2026.
 - [2] Yuxin Cai^{*}, Zongtai Li^{*}, Maonan Wang, **Muyi Bao**, Haokun Zhu, Ruofei Bai, Ding Zhao, Zirui Li, Wenshan Wang, Wei-Yun Yau, Ji Zhang, and Chen Lv. “IntentNav: Learning Spatial-Visual Object Navigation from Human Demonstrations.” *arXiv preprint*, 2026. Submitted to CoRL 2026.
 - [3] **Muyi Bao**, Shuchang Lyu, Zhaoyang Xu, Huiyu Zhou, Jinchang Ren, Shiming Xiang, Xiangtai Li, and Guangliang Cheng. “Vision Mamba in Remote Sensing: A Comprehensive Survey of Techniques, Applications and Outlook.” *Remote Sensing*, 2026.
 - [4] **Muyi Bao**^{*}, Changyu Zeng^{*}, Yifan Wang, Zhengni Yang, Zimu Wang, Guangliang Cheng, Jun Qi, and Wei Wang. “FTCFormer: Fuzzy Token Clustering Transformer for Image Classification.” In *European Conference on Artificial Intelligence (ECAI)*, 2025.
 - [5] Changyu Zeng^{*}, Yifan Wang^{*}, Zimu Wang^{*}, Wei Wang, Zhengni Yang, **Muyi Bao**, Jiming Xiao, Anh Nguyen, and Yutao Yue. “NUMINA: A Natural Understanding Benchmark for Multi-dimensional Intelligence and Numerical Reasoning Abilities.” In *Findings of EMNLP*, 2025.
 - [6] **Muyi Bao**, Shuchang Lyu, Zhaoyang Xu, Qi Zhao, Changyu Zeng, Wenpei Bai, and Guangliang Cheng. “ASP-VMUNet: Atrous Shifted Parallel Vision Mamba U-Net for Skin Lesion Segmentation.” *arXiv preprint*, 2025.
 - [7] **Muyi Bao**, Zeren Tao, Xiaohan Wang, Jiashuo Liu, and Qilei Sun. “Comparative Performance Analysis of Rendering Optimization Methods in Unity Tuanjie Engine, Unity Global and Unreal Engine.” In *IEEE International Conference on Ubiquitous Intelligence and Computing (UIC)*, 2024.
 - [8] **Muyi Bao**, Ming Xu, and Nanlin Jin. “AlexCapsNet: An Integrated Architecture for Image Classification with Background Noise.” *IEEE Access*, 2025.

Skills

-
- **Languages:** Chinese (Native), English (TOEFL iBT: 100/120)
 - **Technical Skills:** Python, PyTorch, Linux, Git, Hugging Face, LoRA/PEFT, VLM fine-tuning, Habitat-Sim, ROS 2, C/C++, MATLAB